PROJECT AND TEAM INFORMATION

Project Title

AIOpsGuard : AI Driven Infrastructure Monitoring, Anomaly Detection, Root Cause Analysis and Intelligent Incident Response Platform

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Organizations in the modern world use servers, applications, and databases to deliver trustworthy digital services to their users. Nowadays, due to the complexity of infrastructure, it has become a hard task to quickly identify malfunctioning system behaviour and understand its cause. High CPU consumption, high memory consumption, application failure, or any unusual logs might result in poorer service quality or interruptions. Different monitoring systems only provide alerts based on preset limits without explaining why the incident took place, whether it will reach critical levels, or what solution can be applied.
This has been illustrated in the respective research papers discussing the application of system metrics and logs for contextual anomaly detection and the application of LLMs for discovering root causes and fixing issues. Based on the above approaches, we would like to present AIOpsGuard, an AI-powered monitoring and intelligent incident management system. The system integrates OS-level resource monitoring, database data, ML for predicting anomalies and failures, and LLMs for finding the cause of problems and providing recommendations. The aim is to change infrastructure management from reactive monitoring to proactive incident prevention so that diagnosis can be made faster and more effectively.

State of the Art / Current solution

However, most of the current solutions still work on a rule-based paradigm and process logs, metrics, and other information independently. As a result, this makes it hard to understand what caused the incident, identify a possible fault in advance, adapt to the changing system state, and conduct necessary corrective actions.
AIOpsGuard will combine these capabilities in one platform. It will monitor operating-system-level resources and processes and employ machine learning to detect anomalies and predict faults. The system will archive historical incidents for future use. The large language model (LLM) will use these data to find root causes and possible preventative measures. For the most critical actions, the system will request approval from a human operator.

Project Goals and Milestones

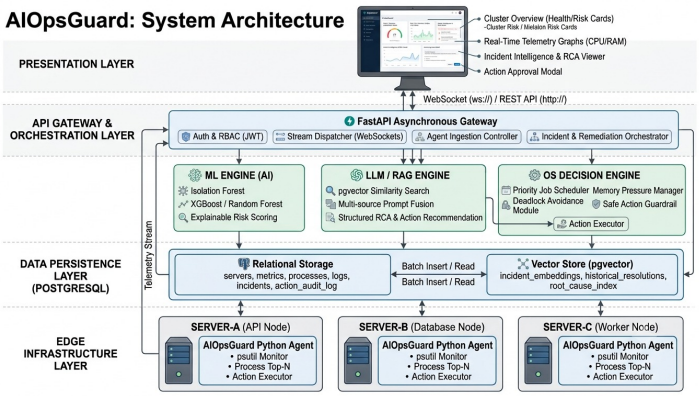
AIOpsGuard's primary aim is to create an innovative platform for continuous infrastructure monitoring and efficient identification of system problems. The system will collect CPU, memory, process, and log data, detect abnormal behaviour, predict potential infrastructure failures, and identify their possible causes.
The major goals and milestones include:

- *Analysis of requirements and definition of monitoring needs.*
- *Database design covering servers, metrics, processes, logs, incidents, and predictions.*
- *Development of OS-level monitoring and resource/process control.*
- *Development of ML-based anomaly detection and failure prediction.*
- *Integration of LLM-based root-cause analysis and incident explanation.*
- *Retrieval of historical incidents and context-based recommendations.*
- *Controlled resource/process management with human approval for critical actions.*
- *Testing and integration of all modules for reliable operation.*

Project Approach

The creation of AIOpsGuard will be in the form of a modular web application based on React.js, FastAPI/Python, and PostgreSQL. The solution will continuously monitor CPU, memory, the number of processes, and system/application logs.
In the context of DBMS, concepts such as ER modeling, normalization, keys, constraints, joins, indexes, views, triggers, and transactions will be used to manage information about the infrastructure, incidents, and historical data.
From the OS perspective, the system will use concepts including thread/process monitoring, CPU scheduling, resource allocation, memory management, synchronization, and deadlock handling.
For Machine Learning, the system will perform anomaly detection and failure-risk prediction. An LLM will analyze metrics and logs to identify root causes, explain incidents, and provide recommendations.
The final solution will be a real-time monitoring dashboard with human approval required for critical actions. The system will be tested by simulating resource overloads, abnormal processes, log anomalies, and recovery scenarios.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

The most prominent result of the project is a working prototype of AIOpsGuard, which is a smart incident-response and infrastructure monitoring platform.
The system will be capable of monitoring servers in real time for CPU and memory usage, processes, operating-system logs, and application logs. It will identify anomalies in server operation and detect possible failures with the help of Machine Learning.
The prototype will include LLM-based root-cause analysis, incident description, and intelligent suggestions based on both historical and current data.
Moreover, the project will deliver a relational database design, database schema, system architecture, OS-level monitoring module, ML model, incident management module, and test results.
The final system will demonstrate how the integration of the operating system, DBMS, ML, and LLM can enable proactive infrastructure monitoring, speed up troubleshooting, and reduce system downtime.

Assumptions

The system assumes that there is a constant availability of server and process monitoring data generated by the monitoring agent during the monitoring process. It also presumes the availability of historical metrics, logs, and incident records for the ML-enabled anomaly detection and prediction. The system presumes internet connectivity for the LLM-powered data analysis. The ML model will first rely on virtual or collected real life data from the infrastructure that meets certain threshold / feature requirements. As for the critical actions of the remediation, they will have to be approved by a system administrator. Initially testing of the system will be carried out in a controlled environment running Linux OS with the help of simulated data on the inaccurate functioning of servers or overload situations of resources.

References

1. *LLM + Cloud Observability. https://doi.org/10.1145/3797161.3797213*
2. *Kubernetes Scheduling. https://doi.org/10.1145/3539606*
3. *ML + Logs/Metrics Anomaly Detection. https://doi.org/10.1109/EDCC.2018.00033*
4. *Academic resources related to Database Management Systems, Operating Systems, Scheduling, and Concurrency Control.*